Review: The real number system

1

a $\{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$

b $\{0, 1, 2, 3, \ldots\}$

c 0

2 The set $\{0, 1, 2, \ldots\}$ is called the set of whole numbers

3

a

i False

ii False

iii True

b

i 38 is a rational, an integer and a whole number.

ii -42 is a rational number and an integer.

iii $\sqrt{6}$ is an irrational number

iv 3.058 is a rational number.

v $\sqrt{49}$ is a rational number, an integer, and a whole number.

vi $\dfrac{\pi}{3}$ is an irrational number.

vii $0.5777\ldots$ is a rational number

viii $\dfrac{26}{45}\ldots$ is a rational number

4 Every rational number can be expressed in the form $\dfrac{p}{q}$ where p is an integer and q is a positive integer.
A whole number can be expressed in this form where $q = 1$ and $p \geq 0$. That is to say, a whole number is a non-negative integer.

5 The decimal could either be a repeating decimal or a terminating decimal.

6

a Whole numbers

b Rational numbers

c Positive real numbers

d All real numbers

e Integers

f Rational numbers

7

a Yes

b No

c No

d Yes

Additional questions

8 The subset of rational numbers includes any number that can be written in the form $\frac{a}{b}$ where a and b are integers and $b \neq 0$, and thus can be written as a terminating or repeating decimal.

9 Possible answers:

- The subset of irrational numbers includes any number that cannot be written in the form $\frac{a}{b}$ where a and b are integers, and thus cannot be written as a terminating or repeating decimal.
- Any negative statement of the answer to question 8 will suffice.

10 No. Irrational numbers can be defined by being not rational and so the two subsets of numbers are mutually exclusive.

11 The integers belong to the subset of rational numbers, since any integer can be written as a fraction with a denominator of 1.

12

a Rational	**b** Irrational	**c** Rational	**d** Rational
e Irrational	**f** Irrational	**g** Rational	**h** Irrational

13

a 4, Rational	**b** $\frac{\pi}{4}$, Irrational	**c** $-\frac{4}{5}$, Rational	**d** 0.08, Rational

14 No. Although the decimal $24.737337333...$ may have a pattern, it cannot be expressed as either a repeating decimal or a terminating decimal, so the number cannot be rational.